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Case Study

Introduction

The expansion of data centers creates a challenge in balancing technological advances, environmental sustainability, and social responsibility. Data centers play a critical role in providing online services and supporting innovation across industries. The benefits of these services are undeniable, but they do not come without a cost. All the while, the industry seems to grow at an unprecedented rate. At the forefront of innovation, the decisions made by data center controlling companies and government regulators guide the future of these facilities. Given the potential to cause negative impacts on the local communities and environment, data center development must stay within ethical bounds. Using a reflexive principlism approach, this case study analyzes the competing stakeholder interests and recommends a balanced solution to guide ethical decision making.

Background

The following section provides background on data center operations, industry growth, and resource demand to establish context for ethical analysis. A data center is a facility that houses computers, storage, and networking systems to support online infrastructure. Inside, servers and related equipment work together to provide computing power for an array of services. Data centers are the foundation of cloud computing, artificial intelligence, and e-commerce. Already making up a huge part of the technological space, these services continue

to expand and innovate at a rapid pace, leading to growth in both their demand and overall impact.

Online services that rely on data centers continue to grow at an unprecedented pace. According to Emma, a cloud management platform, the cloud computing market is valued at $855.7 billion and expanding at a compound annual rate of 18.91% with Amazon, Microsoft, and Google leading the market ("Cloud Market Share Trends"). In the artificial intelligence field, The Guardian reports that OpenAI's chatbot ChatGPT reached 100 million users in its first two months, marking it as one of the fastest-growing consumer internet applications in history (Milmo). Similarly, e-commerce accounted for approximately 16% of total retail sales in the second quarter of 2025 and continues to experience rapid growth, according to the census reports (U.S. Census Bureau). The rise of online services relying on data centers demands careful consideration of environmental and social impacts.

Data centers command large amounts of energy to power the computers and servers inside. Due to the expansion of data centers, the demand and strain on power grids continue to grow. The United States Congress website projects that data center energy consumption could double or triple by 2028, accounting for up to 12% of the U.S. electricity use (Ling). To combat the increase in power consumption on the grid, communities are forced to build more energy infrastructure. Entergy, an energy provider company in the southern region of America, is proposing to build more than $3 billion worth of new gas-fired power plants. Rather than billing Meta directly for these costs, Entergy is proposing to include the costs in rates paid by all customers (Peskoe and Martin). The new energy infrastructure required by data centers is likely to be non-renewable. Per the New York Times, it is forecasted that in 2030, the source of electricity for U.S. data centers will require 450 terawatt-hours of energy. Of that 450

terawatt-hours of power, over 200 terawatt-hours will be supplied from natural gas. Equaling the amount provided by nuclear, solar, and wind combined (Gorelick). Due to the changing power demands of data centers and the imperfect output from the power grid, data centers must use generators to run when the power grid cannot fully support the centers. The generators are commonly powered by diesel gas and release harmful emissions and noise pollution in the process. In California, the top 10% worst diesel emissions zones had a total of 91 centers, while the lower 70% had 47 total centers (Marrinan). Growing power demands strain local power grids and contribute to environmental and social challenges.

In addition to the high energy consumption, data centers require significant amounts of drinking water for cooling. The BBC reports that a typical data center can use between 11 million and 19 million litres of water per day, roughly the same as a town of 30,000 to 50,000 people. While developing its initial AI tools, Microsoft's global water use soared by 34%. During the training of OpenAI's GPT-4, a data center cluster in Iowa used 6% of the district's water supply in one month (Kleinman and Wheeler). In response to growing concerns, Microsoft has announced cooling innovations aimed to reduce or eliminate reliance on drinking water, but these have yet to be implemented (Solomon). These examples expose the heightened water demand caused by data centers, placing pressure on freshwater resources.

In an effort to attract data center investment, local, state, and federal governments have implemented tax breaks on data centers and accelerated federal permitting to incentivize the construction of data centers in their communities. According to CNBC, 42 states provide full or partial sales tax exemptions to data centers or have no state sales tax at all. Of those 42, 37 have passed legislation specifically granting sales tax exemptions for data centers, and 16 of those states have granted nearly $6 billion in exemptions over the past five years (Tortorelli et al.).

These tax incentives are justified by claims that the construction will bring in high-paying tech jobs and long-term economic growth. However, evidence suggests that economic growth and long-term employment may be more limited than anticipated. NPR reports that these centers do generate construction labor for a short period of time and then are maintained by a few hundred jobs at most, not the high-paying tech jobs the government argued the data centers would attract (Bisaha). Additionally, an executive responsible for Microsoft's North American data centers stated in 2024, "I can't think of a site selection or placement decision that was decided on a set of tax incentives," (Nguyen and Green). At the federal level, an executive order has recently accelerated the federal permitting of data center infrastructure, prioritizing rapid development while potentially reducing the depth of review, leaving room for overlooks (Trump). Authorities argue that the incentives and advantages given to huge data center companies boost the local economies, but the actual results may be limited at the cost of the communities.

In response to growing concerns, local and state governments have introduced bills or policies focused on accountability. Although these proposals have not yet passed, they highlight a demand for change in the data center industry. In New York State, Senate Bill S6394A demands full disclosure of water and energy consumption, limiting the use of fossil fuels, and encouraging renewable energy sources. This bill hopes to lower costs for residential consumers impacted by the increased utility costs from data centers ("Senate Bill S6394A"). A similar bill in the state of Georgia, Senate Bill 34, prevents electric utility companies from spreading the costs of data centers' power demand and infrastructure across ratepayers, instead placing the costs on the data center companies ("GA SB34"). While still in the early stages of the legislative process, these proposed bills reflect the increasing need for balance and change in data center regulations.

Method

This case study applies the ethical framework of reflexive principlism to analyze the ethical implications of data centers. Reflexive principlism evaluates decision-making by balancing four core principles: beneficence, nonmaleficence, autonomy, and justice. In this case study, beneficence will be understood as actively doing good. Promoting the well-being, welfare, and best interests of others through action. Nonmaleficence focuses on doing no harm. The active avoidance of causing injury, pain, or unnecessary risk to others. Autonomy emphasizes the freedom to govern oneself. Giving the ability for others to make independent choices, acting according to their own values. Lastly, justice concerns fairness and impartiality. Together, these principles will be balanced to find an ethical solution.

Reflexive principlism is well-suited to this case for several reasons. Stakeholders can be separated into two groups: decision makers and those affected by their decisions. The data center owners and government regulators make up the decision makers, while the surrounding communities and environment represent the patient group or those who bear the consequences of these decisions. Reflexive principlism provides a framework to evaluate all four stakeholders with a single perspective while balancing technical, social, and environmental considerations. Given the rapidly evolving nature of data centers and uncertain potential outcomes, reflexive principlism uses outcome-based and fairness-based considerations to help find the most ethically justified resolution.

Analysis

This case study examines the balance of technological advances and the environmental and social costs of data centers. The stakeholder groups consist of the decision makers, made up of data center owners and government regulators, and the patients, encompassing the

communities and environment surrounding data centers. Tension and conflict arise between decision makers and patients due to the difference in goals and values. To achieve an ethically sound resolution, it is important to weigh all stakeholders and their values equally. Using reflexive principlism, the study will evaluate how each principle, beneficence, nonmaleficence, autonomy, and justice, applies to each stakeholder and what it means to them, how it is currently displayed, and where gaps or shortcomings exist.

The first stakeholder to analyze is the data center owners. Beneficence looks like positively impacting the communities they operate in through economic growth, providing reliable and impactful technological infrastructure, and finding solutions that benefit the environment. Currently, data center owners display some of these by supporting various online services, such as cloud computing, that continue to benefit and advance individuals and businesses. These centers are increasingly more accessible, reliable, and efficient, reflecting ongoing efforts of technological innovation. The centers also provide construction and maintenance jobs to communities with few opportunities. However, there is much to improve upon in this area. Although local communities do receive some economic gain, it is often for a short period of time, far too small, and outweighed by increased water and energy consumption. The benefits of data centers are more directed to global and corporate groups than to local communities. Data center owners often overlook the environmental and social consequences of increased water and power demand, prioritizing revenue and profit.

For data center owners, nonmaleficence entails minimizing water and power consumption in local communities, reducing power obtained from nonrenewable sources and diesel generators, and avoiding strain on local infrastructure. Currently, some nonmaleficent practices are evident, such as advances in cooling technologies, reducing the need for water, and

complying with existing environmental regulations. However, the data center owners continue to rely on fossil fuels and diesel generators for energy. The increasing demand for water and power necessitates infrastructure investments. The costs continue to be passed on to local communities rather than the data center owners. While there's some effort to reduce harm, negative impacts on communities and the environment continue to persist.

For data center owners, justice involves the fair distribution of costs to those who profit from data centers, transparent water and power consumption, and supporting required infrastructure upgrades. Currently, fairness is displayed through the payment of some local taxes and limited transparency of resource consumption. Justice could be improved by preventing the shifting of costs to local residents, maximizing transparency of the impact of data centers on local communities' resources and costs, and supporting the required upgrades to keep the data centers functioning.

Autonomy looks like freedom to make business decisions on land and facilities they own, and the capacity to innovate and respond to market demand. This principle is currently shown through their ability to independently manage data centers and voluntary sustainability commitments and self-regulation. Data centers are still subject to permitting processes and regulatory requirements.

The second stakeholder to analyze is the government and regulators. Under the principle of beneficence, their ethical responsibility includes promoting economic growth to local and global communities through appropriate rules and regulations while valuing long-term economic and environmental sustainability. This is currently reflected through tax incentives to data center projects in hopes of local and global economic gain, as well as proposed bills focused on positive long-term environmental impact. Beneficence is lacking due to limited evidence that tax

incentives to data centers create a positive and long-term impact on local communities. Government regulators must continue to support regulations that support future economic growth and the protection of environmental resources.

Under the principle of nonmaleficence, government regulators should prevent harm to the local communities and environment, including minimizing strain on power grids and water systems. This responsibility is currently reflected through the lengthy permitting process of data centers, which considers zoning, land use, and environmental impact, as well as limited emission and efficiency regulations set in place. Nonmaleficence is compromised by the acceleration of the data center permitting process, creating the opportunity for some aspects to be overlooked or skipped. Additionally, a lack of expanded regulations to limit the use of fossil fuels and diesel generators for data centers leaves communities and ecosystems vulnerable to the negative impacts of data centers.

For government regulators, justice entails a required distribution of costs to the appropriate parties, preventing utility cost shifting to residents, and giving power to local communities to support their own interests. Justice is reflected through proposed bills to combat cost shifting and protect local communities from additional costs, as well as the opportunity for local governments to pass bills aligned with their needs. To fully uphold justice, governments should prioritize passing legislation supporting local communities and governments at the federal level rather than yielding primarily to the interests of data center operators.

Under the principle of autonomy, regulators have a responsibility to allow data centers to act in their best interest as long as regulations and rules are followed, while supporting local communities to participate in decision-making. This principle is currently reflected through the operational freedom given to data centers and the ability of local governments to propose bills in

the interest of the communities. Strengthening mechanisms for local participation and oversight would better support autonomy.

The next stakeholder group, local communities, is a part of the patient group and thus not directly involved in the decision-making processes of data centers. Instead, their focus is on how data centers impact their lives. In terms of beneficence, ethical responsibility involves providing long-term economic opportunity and positive infrastructure investment into the community from data centers. Currently, beneficence is displayed through the short-term benefit of construction labor and limited long-term opportunities. The beneficence principle is lacking in large-scale, long-term, high-paying job opportunities, and the infrastructure development associated with data centers often fails to benefit the local community.

The principle of nonmaleficence in local communities comprises the protection from the negative effects of data centers, including increased demand for water, power, and land. Valuing protection from higher costs, air and noise pollution, and water shortages. Currently, some protection is provided through limited environmental and permitting regulations. Nonmaleficence is lacking because communities continue to be exposed to diesel emissions from backup generators, higher costs from increased power demand, and a lack of fresh water.

For local communities, justice looks like a fair distribution of costs and benefits and meaningful opportunities to influence decisions made by decision makers. Justice is currently partially reflected through the recognition of unfair cost shifting. Justice is lacking because residents continue to be billed the increased utility rates due to the heightened demand for energy and water, and the lack of local governments' influence in decision-making.

Under the principle of autonomy, local communities have the ethical expectation to influence decisions regarding data center permits, zoning, and regulations, as well as having

access to clear information about the potential impacts of data centers on the local community. Some autonomy is reflected through public hearings and limited influence on regulations. Autonomy remains limited in practice. Communities continue to lack access to useful information about the environmental, economic, and social impacts of data centers. Local communities still possess limited power to affect the decisions being made in their own communities.

The final stakeholder group is the environment. The principle of beneficence entails active restoration of ecosystems affected by data center construction, investing in renewable energy sources, and reducing carbon and water footprints. Some beneficial practices are reflected in proposed bills limiting fossil fuel consumption and corporate sustainability pledges. Beneficence is lacking due to the continued use and expansion of nonrenewable energy sources, and environmental sustainability efforts are limited to pledges or proposed regulations.

Under the principle of nonmaleficence, ethical responsibility entails avoiding environmental damage, minimizing water shortages, and reducing diesel emissions. Some progress is evident through limited efficiency improvements of power and water systems. Data centers continue to rely on fossil fuels for power, drinking water for cooling, and the use of diesel generators, contributing to long-term environmental harm.

Justice for the environment demands the sharing of natural resources like power and water, and intergenerational fairness to future generations. Currently, justice is displayed through proposed policies on sustainability and regulating resource usage. Justice remains limited due to the lack of significant sustainability efforts, environmental harms treated as acceptable trade-offs by data center owners, and the continued large consumption of energy and power.

Under the principle of autonomy, the environment is entitled to respect and self-sufficiency through protective regulations. Some acknowledgement of environmental autonomy is evident through limited environmental regulations and proposed environmental legislation. Environmental autonomy is still lacking because economic interests continue to outweigh environmental interests. Regulations supporting the environment are limited or remain in the legislative process. This imbalance calls for stronger, more enforceable measures to protect the environment.

Recommendation

This section proposes three unique solutions and recommends one based on stakeholder analysis. The first possible solution is to maintain the status quo, preserving the current data center industry environment without additional regulations. From a beneficence perspective, this approach supports technological advancements, global economic growth, and reliable and powerful digital infrastructure. It allows data centers to continue to expand and scale rapidly. However, when viewed from a nonmaleficence perspective, this solution continues to harm local communities and the environment through increased energy and water strain, emissions, and cost shifting to local communities. From a justice perspective, cost-shifting persists, and decision-making power remains unbalanced among data center owners. This solution favors the autonomy of data center owners because rules and regulations will continue to preserve their freedom to operate. The autonomy of communities and the environment remains limited due to their minimal influence over decisions that affect them. Overall, maintaining the status quo favors the decision makers and technological innovation while failing to protect the local communities and environment.

Another potential solution is the enforcement of moderate regulatory measures. These regulations would focus on energy efficiency, reducing water consumption, and limiting emissions, while requiring partial reporting and transparency of resources consumed and limited community input on decisions. From a beneficence perspective, this approach hopes to balance technological growth with improved support for local communities through transparency and efficiency. This solution supports nonmaleficence by limiting harm to communities and still allowing the data centers to operate in their best interests. From a justice perspective, this solution improves fairness by limiting cost-shifting practices and granting communities a greater, though still limited, voice in decision-making. Regarding autonomy, this approach slightly constrains the freedom of data centers through additional regulations, but expands the autonomy of local communities in the process. Government regulators retain autonomy through existing legislative and regulatory processes. Overall, moderate regulation improves ethical balance without significantly impeding innovation.

The final and recommended option is the implementation of heavy regulations. This approach would enforce strict environmental and resource regulations, require full transparency in energy and water consumption, mandate community input on decision-making, ban cost shifting, and heavily incentivize renewable energy use. This solution supports beneficence by maximizing the positive impact on communities and the environment by prioritizing sustainable and renewable practices. In terms of nonmaleficence, this approach offers the strongest protection against future harm to the environment and communities by minimizing environmental degradation, overuse of resources, and public health risks. When implemented properly, both decision makers and patients can avoid unnecessary harm. Under the principle of justice, this solution ensures fair distribution of costs by banning cost shifting and giving a voice

to local communities and the environment. Although autonomy for data center owners is reduced, this is ethically justified given the huge increase in autonomy towards local communities and the environment. This solution still allows data centers to act independently. While heavy regulation may create higher short-term costs or limit innovation for data center owners, it supports long-term sustainability, protects future generations, and encourages economic growth through renewable energy and infrastructure development. As a result, this approach best aligns with the ethical framework of reflexive principlism.

Conclusion

The expansion of data centers creates ethical tension between technological innovation and environmental and social responsibility. Using reflexive principles, this case study examined the competing interests of data center owners, government regulators, local communities, and the environment. This analysis shows that although data centers provide significant benefits, the burden is disproportionately placed on local communities and the environment. The proposed solution of heavy regulations best balances beneficence, nonmaleficence, justice, and autonomy by reducing harm, ensuring fair cost distribution, increasing transparency, and improving community input, while still allowing data centers to operate independently. Although this approach limits data centers, it supports long-term sustainability and technological progress.

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